

Streamlining Physiological Observations in Immersive Virtual Reality Studies with the Virtual Reality Scientific Toolkit

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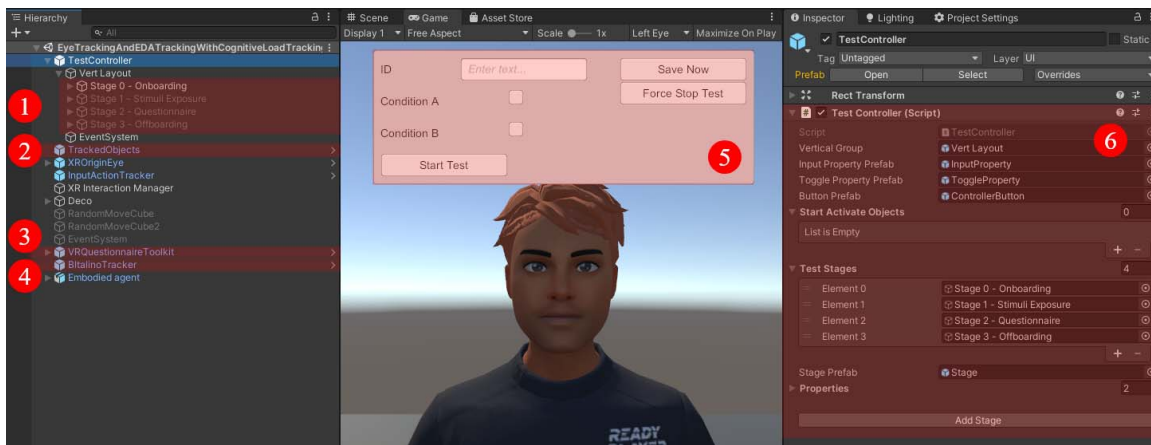


Figure 1: Experimentier view during a study utilizing the Virtual Reality Scientific Toolkit.

1. List of Stages, 2. List of Tracked Elements, 3. In-VR Questionnaire, 4. EDA Tracking, 5. Experimentier Interface, 6. Stage Setup

ABSTRACT

The open-source *Virtual Reality Scientific Toolkit* (VRSTK) facilitates designing and conducting user studies in immersive virtual reality (VR) environments within the Unity 3D real-time development platform. Features include recording and reviewing scenes, collecting and evaluating user-specific data from sensors or in-VR questionnaires, and remote and multi-user setups. In this paper, we describe the core features of the VRSTK by guiding the reader through a hypothetical user study.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies

1 INTRODUCTION

Several challenges need to be overcome when preparing for and conducting a study in immersive virtual reality. In addition to basic knowledge of the principles of real-time development platforms (e.g. Unreal Engine or Unity3D), test routines have to be implemented. These functions range from data acquisition, recording, storage, and synchronization. An overview of features offered in toolkits with a specific focus on human factors, including the VRSTK, is given in [10].

Since many of these routines are common to a wide range of experimental designs, it is preferable to rely on templates and toolkits

that have already been tested and verified by others, rather than reimplementing them over and over again. This will not only speed up the development process but also reduce errors.

The Virtual Reality Scientific Toolkit provides such a framework. It facilitates the creation and execution of human-centered experiments in immersive VR environments, simplifying their development and reducing redundant tasks. The VRSTK is an open-source toolkit that extends the capabilities of Unity and is released under the MIT license. Unity was chosen as the underlying platform because it is a commonly used tool in VR research and offers a comparatively easy entry into developing custom 3D applications. In addition, Unity comes with an Asset Store that provides a variety of helpful scripts and 3D objects.

Features of the VRSTK include recording and reviewing experiments, collecting and analyzing user-specific data from sensory information provided by the headset or additional hardware, or in-VR questionnaires. The VRSTK can also support multiplayer applications by using the open-source netcode for Unity Mirror¹. This allows a) the experimenter to be in the same virtual room even though the participant and experimenter are in different locations; b) to run an experiment with more than just one participant at a time; c) to run experiments where participants must interact with each other; and d) to take advantage of concepts such as asymmetric normalization as proposed in [2].

Another critical aspect of immersive VR experiments is the way participants are embodied. As studies have shown, there are a large number of different avatars used in VR multiplayer applications. These avatars vary in fidelity and representation characteristics and thus influence the way others are perceived [6]. Because the avatars provided by Ready Player Me² are used in many applications³, are easy to customize, and offer blend shapes to provide a rich form

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¹<https://github.com/MirrorNetworking/Mirror>

²<https://readyplayer.me/>

³<https://readyplayer.me/apps>

of facial expression, the VRSTK integrates the Ready Player Me avatar service. Researchers are enabled to use their own, customized avatars for multi-user studies.

The VRSTK supports a variety of head-mounted displays (HMDs) through the generic OpenXR⁴ standard. Following the current state of Unity VR development, the VRSTK uses the VRInteraction-Toolkit⁵ for locomotion and interaction.

In addition to basic features such as object tracking within the virtual environment, more sophisticated features and analysis can be accessed using different sensors. The following sensors are already implemented:

- **Electroencephalogram (EEG)** using Epoc X⁶ by Emotiv
- **Electrodermal Activity (EDA)** using BITalino⁷ by Plux Biosignals
- **Electrocardiography (ECG)** using BITalino by Plux Biosignals
- **Physical Object Tracking** using HTC Vive Trackers⁸ by HTC
- **Full Body Tracking** using HTC Vive Trackers by HTC
- **Gaze Tracking** as featured in HTC Vive Pro Eye⁹ by HTC
- **A generic serial interface** that facilitates the integration of a wide range of sensors such as humidity or temperature sensors from platforms like Arduino

Most features are implemented in a modular fashion and can be distributed with specific, predefined scenes. Depending on the specific needs of the experimenter, these predefined scenes can be modified and combined. For each participant, the entire study process can be recorded and stored. The ability to simultaneously collect data from multiple sensors and review it using the replay function allows the experimenter to better understand participants' physiological and cognitive responses to stimuli, thus facilitating more accurate and reliable results. All tracked features can be exported in the JSON file format for further data analysis in Python, R, or any other statistics software.

In addition to a low barrier to entry and less effort for the general setup, the VRSTK aims to increase the comparability and thus the replicability of studies conducted in immersive VR [5].

2 EXEMPLARY DEPLOYMENT

In the following, we demonstrate the utilization of the VRSTK on the simple example of setting up a hypothetical between-group study on human gaze interaction with an embodied agent, under two not further described conditions. Participants shall be physiologically monitored during the study, meaning the tracking of their gaze direction, movement, as well as EDA data. We thereby assume the agent is an already developed entity, employable via Unity's custom package system.

⁴<https://www.khronos.org/OpenXR/>

⁵<https://github.com/Unity-Technologies/XR-Interaction-Toolkit-Examples>

⁶<https://www.emotiv.com/epoc-x/>

⁷<https://www.bitalino.com/storage/uploads/media/sensor-datasheet-psychobit-revb-2.pdf>

⁸<https://www.vive.com/us/accessory/tracker3/>

⁹<https://www.vive.com/us/product/vive-pro-eye/overview/>

Table 1: Stage descriptions of the exemplary study.

Stage	Name	Task	Monitoring
0	Onboarding	Consent form and description of the upcoming sequence of events.	
↓	Manual Transition		
1	Stimuli Exposure	Virtual interaction with an agent under assigned condition.	X
↓	Transition after 50s		
2	Questionnaire	Subjective evaluation of the agent interaction.	X
↓	Manual Transition		
3	Offboarding	Acquisition of demographic data and further course of action.	
∞	Manual Ending		

2.1 Setup

In order to set up the basic test environment, a quickstart guide is available on the VRSTK GitHub repository¹⁰. The newest releases of the VRSTK are regularly published on the repository as well. After importing the .unity package and other dependencies in Unity as described in the documentation, various pre-defined sample scenes can be found. For our exemplary study, we start using the GazeTracking scene, since it already contains the prerequisites for tracking of gaze data. Extending this template scene, the virtual study environment can be built or inserted using common Unity development. In the current example, this means adding the embodied agent and agent-behavior-related scripts to the scene as well as creating a virtual test environment.

Each example scene contains a `TestController` object which provides a way to structure the study in different stages. For the current example, we want to establish four study stages, as described in Table 1, and thus, add them in the `TestController` (see Item 6 in Fig. 1).

For each of the four different stages, the experimenter can define custom menus containing study procedures such as time or event based transitions between those stages. Stages added via the `TestController` appear in the hierarchy (Item 1 in Fig. 1) and contain further individual adjustments as seen in Fig. 2. In the current example, stage 0 can be manually started by the experimenter. The transition to stage 1 is also handled manually by the experimenter, whereupon the time-synchronized tracking of the player as well as the gaze and EDA sensors are activated by adding them to the stage-specific `GameObjects to activate` list (see Fig. 2). A timed transition ends the stimuli exposure phase after 50s and leads to stage 2 automatically. Tracking of physiological features is still active during the questionnaire evaluation in stage 2 in order to determine conscientious completion. It is disabled via the `GameObjects to deactivate` list (also seen in Fig. 2) with the transition to stage 3 where demographic data is acquired and the participants are debriefed. Finally, as a participant finishes stage 3, a JSON file of the tracked data is saved.

By using the `TestController` an experimenter interface can be generated as shown in Fig. 1 Item 5. It allows the real-time control of the study like switching stages, adding custom notes, adjusting conditions, or triggering events. If desired, this input can also be recorded.

The main objective of our hypothetical study is the observation of gaze depending on the agent's behavior. Some prerequisites for the data collection during task performance in stage 1 and the questionnaire in stage 2 are already configured in the template scene (i.e. eye movement and name of the object that was focussed). Additionally, the agent's movement is recorded. For this purpose,

¹⁰<https://github.com/ixperience-lab/VRSTK>

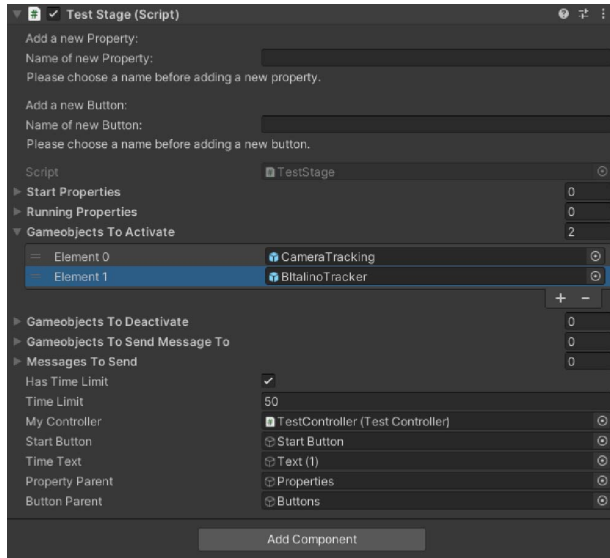


Figure 2: Exemplary Setup for stage 1. Objects and tracking elements in the scene can be activated or deactivated during stage-transitions by adding them to the according list.

the VRSTK provides a menu-item within the Unity editor's menu bar to add variables like position and rotation of in-game objects to the data collection (Fig. 3). All tracked elements are summarized in a list contained in the TrackedObjects object (see Item 2 in Fig. 1).

The fusion of the eye-tracking and EDA sensors allows a more stable and accurate observation of a participant's cognitive load during stimuli exposure in stage 1 and the questionnaire stage 2 (see [1,8]).

With the Vive Pro Eye HMD, the VRSTK supports an eye tracking solution enabling researchers to utilize pupillometry and to categorize fixations by the object in focus.

To enable EDA tracking the BITalino (Item 4 in Fig. 1) prefab must be added to the current scene environment. In order to receive data from the EDA sensor, the BITalino software is needed¹¹. For the communication between the BITalino software and Unity, we decided to use the LabStreamingLayer (LSL)¹² and the OpenSoundControl (OSC)¹³ protocols. The data reception progress runs on a separate thread within Unity in order to enable tracking in a higher sample rate (A detailed description for the EDA setup can be found in the documentation).

Research has shown that integrating a questionnaire directly within VR instead of presenting it afterwards in the real-world offers several advantages such as less spatial disorientation [7] and reduced systematic bias due to eventual presence-loss [9]. Therefore, the VRSTK integrates and extends the open-source VRQuestionnaireToolkit [4].

The questionnaire can be set up via the *Generate Questionnaire* option in the VRSTK menu-bar. This adds a questionnaire object to the scene, as seen in Item 3 in Fig. 1. Common questionnaires like NASA TLX, Simulation Sickness Questionnaire, IPQ and SUS Presence Questionnaire, or the System Usability Scale are already included in the VRQuestionnaireToolkit package. By adding it to the stage-specific GameObjects to de/activate list, the questionnaire is shown in stages 2 and 3.

¹¹<https://support.pluxbiosignals.com/knowledge-base/introducing-opensignals-revolution/>

¹²<https://github.com/scen/labstreaminglayer>

¹³<https://ccrma.stanford.edu/groups/osc/index.html>

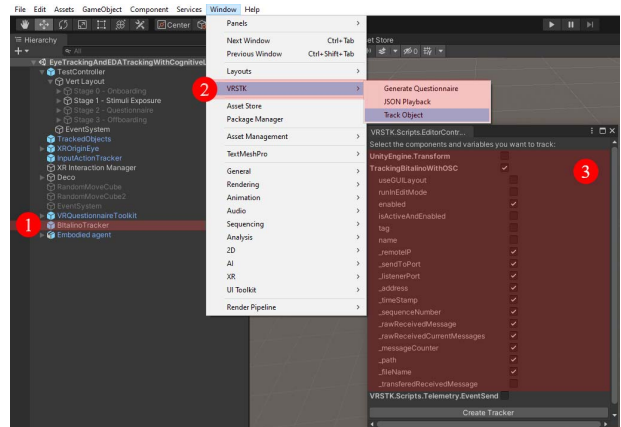


Figure 3: Various variables can be marked to be tracked via the menu bar. 1. Select object to track, 2. Select *Track Object*, 3. Choose desired variables

2.2 Execution

An in-VR study utilizing the VRSTK runs directly out of the Unity editor, enabling the experimenter to observe and control the study from a regular computer screen, while the participant puts on a connected desktop-VR HMD. The study starts as soon as the experimenter activates the first stage. Following stages are switched on one after another, as previously described in a time-controlled manner (automatically) or manually. In the current example, the transition between the stages is as shown in Table 1.

Thereby, the experimenter can stop and intervene at any time (stop and save). After the last stage is stopped, all tracked items are saved with their selected properties as a JSON file.

2.3 Replay and Analysis

After the study has been carried out, the VRSTK supports researchers in subsequent data analysis. For example, the stored data can be replayed locally, which enables subjective analysis and facilitates data transparency. Objective analysis is partly streamlined with the VRSTK's provision of R and Python scripts with a wide range of functionalities.

2.3.1 Replay

A saved study can be replayed for a more detailed analysis by loading and playing the stored file in the Replay window (Fig. 4). Replay is thereby restricted by the configured sampling rates during the recording. As with a common media player, the experimenter has the ability to control the playback of the loaded file. Furthermore, for a more detailed analysis, the experimenter can view the scene from different perspectives (e.g., embodied agent perspective).

Applied to the example study, feedback from the tracked sensors is given in stage 1 and 2. For example, as shown in Fig. 5, eye tracking shows a participant's movement as well as a ray for gaze direction and fixations with spheres. The pupil values, as well as the tracked EDA sensor values are displayed in the Unity inspector, which are then processed in the subsequent analysis.

2.3.2 Analysis

The VRSTK provides further data processing templates such as scripts to convert the stored data into structures specific to analysis-languages such as R and Python for further evaluation. For example, when different sensor systems are captured, this often leads to a recording with different sampling rates. To synchronize the sensor signals, data must be sampled up or down, which is facilitated by

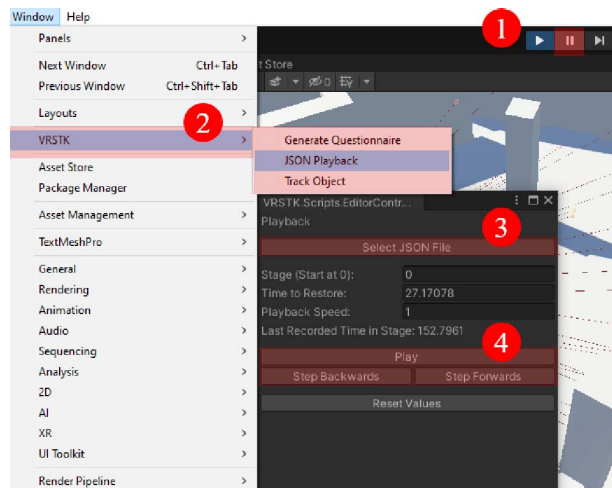


Figure 4: Similar to a generic video player, recorded participants data can be played back via the *Replay* feature of the VRSTK. 1. Enter Unity's playmode and pause, 2. Select JSON Playback, 3. Select recorded participant data, 4. Play

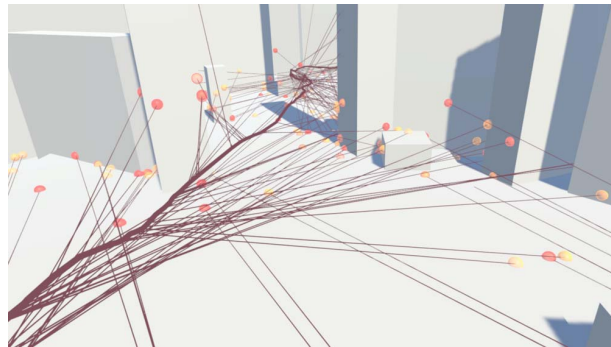


Figure 5: Replay of recorded gaze, and fixations during a participants level exploration.

scripts supplied with the VRSTK. Furthermore, the VRSTK also provides visualizations in R that are useful for subjective observation such as heatmaps for eye tracking.

3 OUTLOOK

With the release of the VRSTK, v1.0.0, we hope to support the research community by providing a tool that makes it more convenient and easier to conduct studies in immersive VR. The VRSTK can be used free of charge and the source code can be assessed and modified.

Different members of our lab have conducted several user studies with the VRSTK. With each new study, the research group has gained more knowledge about what features are needed and how to improve the toolkit. For example, when using the questionnaire, it is more difficult and time-consuming to enter a free-text response with a virtual keyboard than with a regular keyboard. Therefore, we plan to implement tracked, real keyboards in the virtual environment, as described by Meta recently¹⁴.

The highest priority for implementation are those features that can be considered the lowest common denominator across studies.

¹⁴<https://www.meta.com/help/quest/articles/headsets-and-accessories/meta-quest-accessories/tracked-keyboards-meta-quest/>

With the recent trend towards standalone HMDs such as the Meta Quest family, the next step is to make these headsets more integrated, e.g. by providing an experimenter screen within a second HMD that was originally displayed on a 2D monitor connected to the participant's HMD. The benefits of this approach, such as a collective onboarding for many participants, have already been observed in a first version providing these features in the *Shared Excellence - Laboratory Learning Spaces 4.0* [3] project.

With this article, in addition to describing the toolkit, we hope to inspire other scientists and interested developers to use and contribute to the VRSTK.

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REFERENCES

- [1] I. Brishtel, S. Ishimaru, O. Augereau, K. Kise, and A. Dengel. Assessing cognitive workload on printed and electronic media using eye-tracker and EDA wristband. In *Proceedings of the 23rd International Conference on Intelligent User Interfaces Companion*. ACM, Mar. 2018. doi: 10.1145/3180308.3180354
- [2] J. Deuchler, D. Hepperle, and M. Wölfel. Asymmetric normalization in social virtual reality studies. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, Mar. 2022. doi: 10.1109/vrw55335.2022.00019
- [3] J. Deuchler and M. Wölfel. Lessons learned in transferring a lecture on virtual reality into immersive virtual reality. In P. A. Henning, M. Striwe, and M. Wölfel, eds., *20. Fachtagung Bildungstechnologien (DELFI)*, pp. 117–122. Gesellschaft für Informatik e.V., Bonn, 2022. doi: 10.18420/delfi2022-022
- [4] M. Feick, N. Kleer, A. Tang, and A. Krüger. The virtual reality questionnaire toolkit. *UIST '20 Adjunct*. ACM, 2020. doi: 10.1145/3379350.3416188
- [5] D. Hepperle, T. Dienlin, and M. Wölfel. Reducing the human factor in virtual reality research to increase reproducibility and replicability. *2021 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 100–105, 2021.
- [6] D. Hepperle, C. F. Purps, J. Deuchler, and M. Wölfel. Aspects of visual avatar appearance: self-representation, display type, and uncanny valley. *The Visual Computer*, 38(4):1227–1244, June 2021. doi: 10.1007/s00371-021-02151-0
- [7] J. Knibbe, J. Schjerlund, M. Petraeus, and K. Hornbæk. The dream is collapsing: The experience of exiting vr. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, CHI '18, p. 1–13. ACM, 2018. doi: 10.1145/3173574.3174057
- [8] L. Liberman and I. Dubovi. The effect of the modality principle to support learning with virtual reality: An eye-tracking and electrodermal activity study. *Journal of Computer Assisted Learning*, Nov. 2022. doi: 10.1111/jcal.12763
- [9] S. Putze, D. Alexandrovsky, F. Putze, S. Höffner, J. D. Smeddinck, and R. Malaka. Breaking the experience: Effects of questionnaires in vr user studies. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, CHI '20, p. 1–15. ACM, 2020. doi: 10.1145/3313831.3376144
- [10] M. Wölfel, D. Hepperle, C. F. Purps, J. Deuchler, and W. Hettmann. Entering a new dimension in virtual reality research: An overview of existing toolkits, their features and challenges. In *2021 International Conference on Cyberworlds (CW)*. IEEE, Sept. 2021. doi: 10.1109/cw52790.2021.00038